

Why energy system modelling?

- **Energy system modelling** is about forming an understanding of the design and operation of energy systems.
- **Complex, many interacting parts:**
 - renewables differ in space and time
 - transmission grid constraints
 - storage and demand-side management
 - electrification of other sectors
- **Need a lot of data** to model the system.

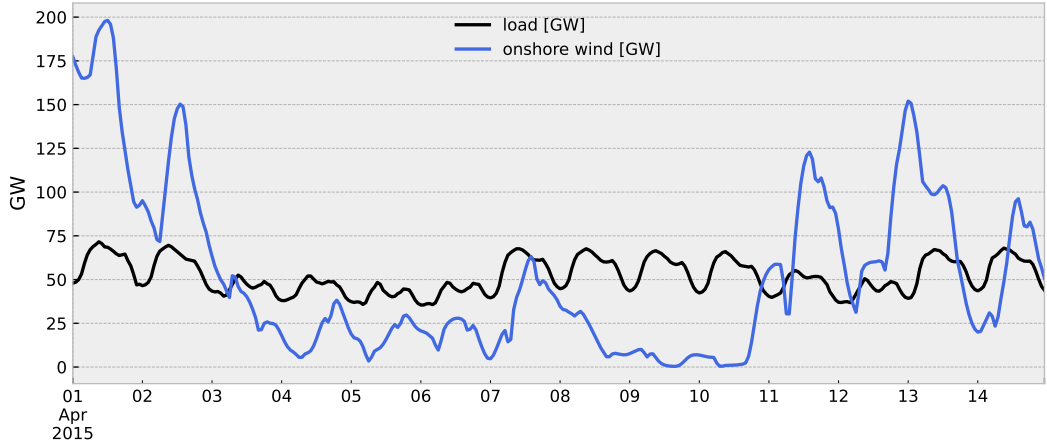
Today – Current Energy Systems

- ensure stable/reliable system operation
- study impacts of price shocks
- study impacts of policy changes

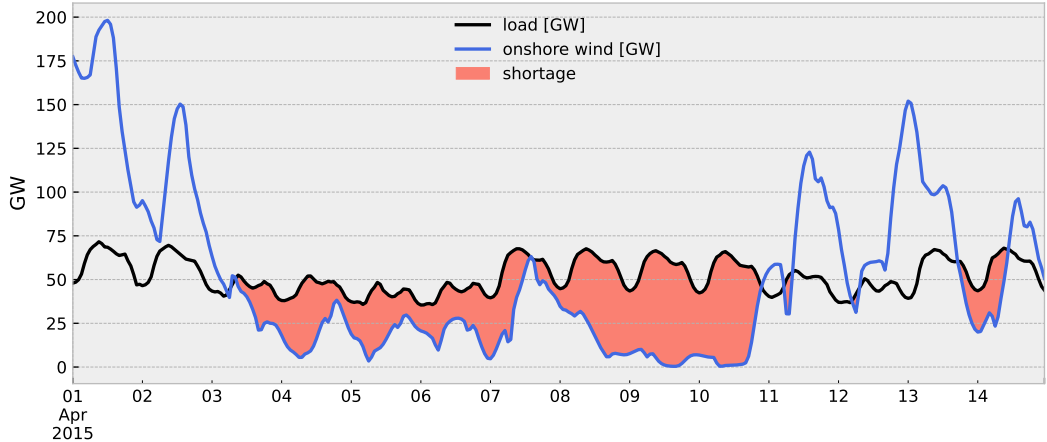
Tomorrow – Future Energy Systems

- trial new market designs in simulations
- transition to climate-neutrality
- infrastructure for renewables integration
- system operation with many renewables

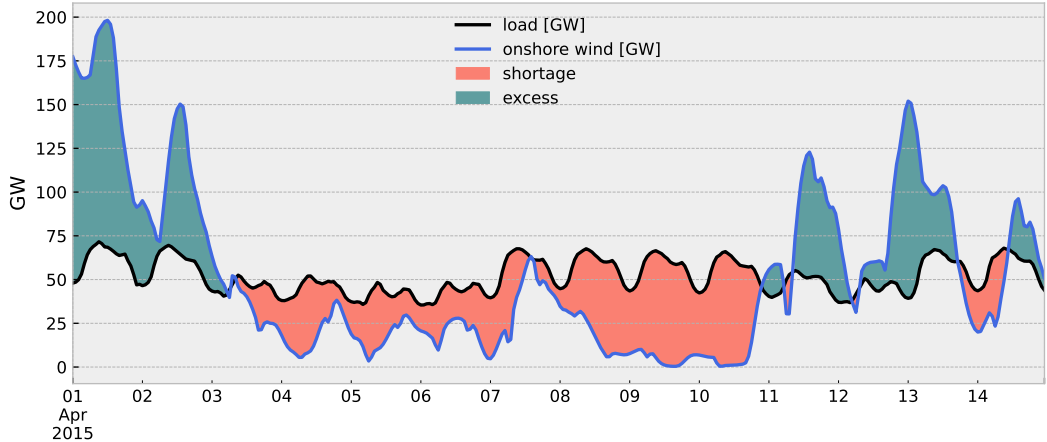
Temporal Challenges: Balancing generation with demand



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Cost-optimal system with 100% renewable energy?

Build your own zero-emission electricity supply

Introduction

This tool calculates the cost of meeting a constant electricity demand from a combination of wind power, solar power and storage for different regions of the world.

First choose your location to determine the weather data for the wind and solar generation. Then choose your cost and technology assumptions to find the solution with least cost. Storage options are batteries and hydrogen from electrolysis of water.

Fun things to try out:

- remove technologies with the checkboxes, e.g. hydrogen gas storage or wind, and see system costs rise
- set solar or battery costs very low, to simulate breakthroughs in manufacturing

See also this [Twitter thread](#) for an overview of the model's features and capabilities.

This is a **toy model** with a **strongly simplified** setup. Please read the [warnings](#) below.

Step 1: Select location and weather year

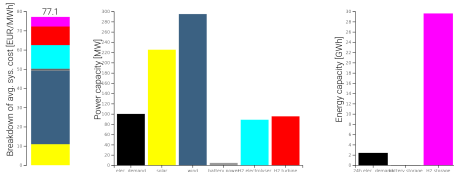
- ☒ Select country
- ☐ Select state or province (for Australia, China, Germany, India, Russia and United States)
- ☐ Select point, rectangle or polygon, using the toolbox that appears at top-right



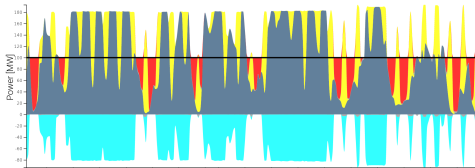
Average system cost [EUR/MWh]: 77.1

Cost is per unit of energy delivered in 2015 euros. For comparison, household electricity rates (including taxes and grid charges) averaged 211 EUR/MWh in the [European Union in 2018](#) & 132 USD/MWh in the [United States in 2019](#)

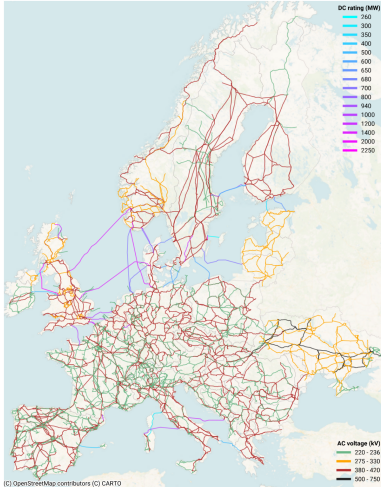
Average marginal price of hydrogen [EUR/MWh LHV]: 78.8, [EUR/kg]: 2.60



Dispatch over a year to meet the constant demand (electricity demand is the black line; you can zoom and pan to see the details; negative values correspond to storage consuming electricity):



Spatial Challenges: Balancing subject to grid congestion



- Best renewable resources may not be close to demand centres.
- But can only transport electricity within **restrictions**:
- Must respect **Kirchhoff's laws**.
- Must respect **thermal limits**.
- Other networks? Gas, H₂, CO₂?

PyPSA:

Python for Power System Analysis

Capabilities

Capacity expansion (linear)

- single-horizon
- multi-horizon

Market modelling (linear)

- Linear optimal power flow
- N-1 security-constraints
- Unit commitment
- Redispatch
- Rolling-horizon

Non-linear power flow

Modelling-to-Generate-Alternatives

With components for

- Electricity transmission networks and pipelines
- Generators with **unit commitment constraints**
- **Variable** generation with time series (e.g. wind, solar, hydro)
- **Storage** with efficiency losses and inflow/spillage for hydro
- **Conversion** between energy carriers (PtX, CHP, BEV, DAC)

Backend

- all data stored in **pandas**
- framework built for performance with large networks and time series
- interfaces to major **solvers** (Gurobi, CPLEX, HiGHS, Xpress), with **linopy** (by PyPSA devs)
- highly **customisable**, but **no GUI**
- Suitable for greenfield, brownfield & pathway studies



The plan

1 **Morning:** Modelling of electricity systems with PyPSA.

- Introduction and short-term dispatch optimisation
- Capacity expansion modelling for long-term planning
- **Hands-on exploration**

2 **Afternoon:** Modelling of sector-coupled systems with PyPSA.

- Modelling of sector-coupled systems (heating, hydrogen, electric vehicles)
- **Hands-on exploration**
- Overview of the European energy system model PyPSA-Eur.

Website: <https://fneum.github.io/nordnet/>